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(54) **PLASMA PROCESSING APPARATUS AND
PLASMA PROCESSING METHOD**

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(58) **Field of Classification Search**
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See application file for complete search history.

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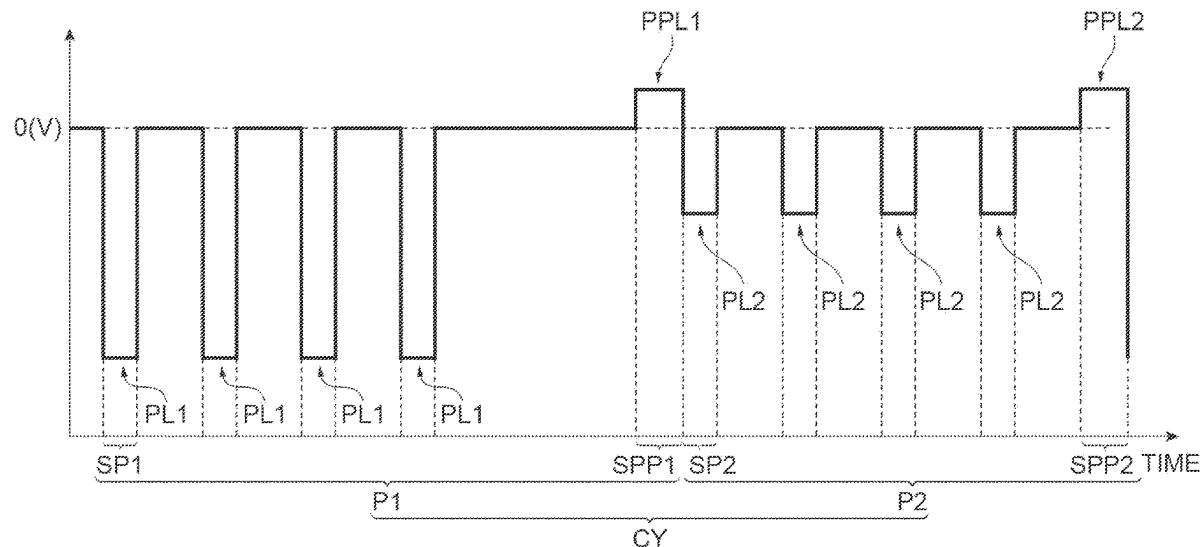
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(57) **ABSTRACT**

The disclosed plasma processing apparatus is provided with a chamber, a substrate support, and a power source system. The substrate support has an electrode and configured to support a substrate in the chamber. The power source system is electrically connected to the electrode and configured to apply a bias voltage to the electrode to draw ions from a plasma in the chamber into the substrate on the substrate support. The power source system is configured to output a first pulse to the electrode in a first period and output a second pulse to the electrode in a second period after the first period, as the bias voltage. Each of the first pulse and the second pulse is a pulse of a voltage. A voltage level of the first pulse is different from a voltage level of the second pulse.

12 Claims, 9 Drawing Sheets



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Fig. 1

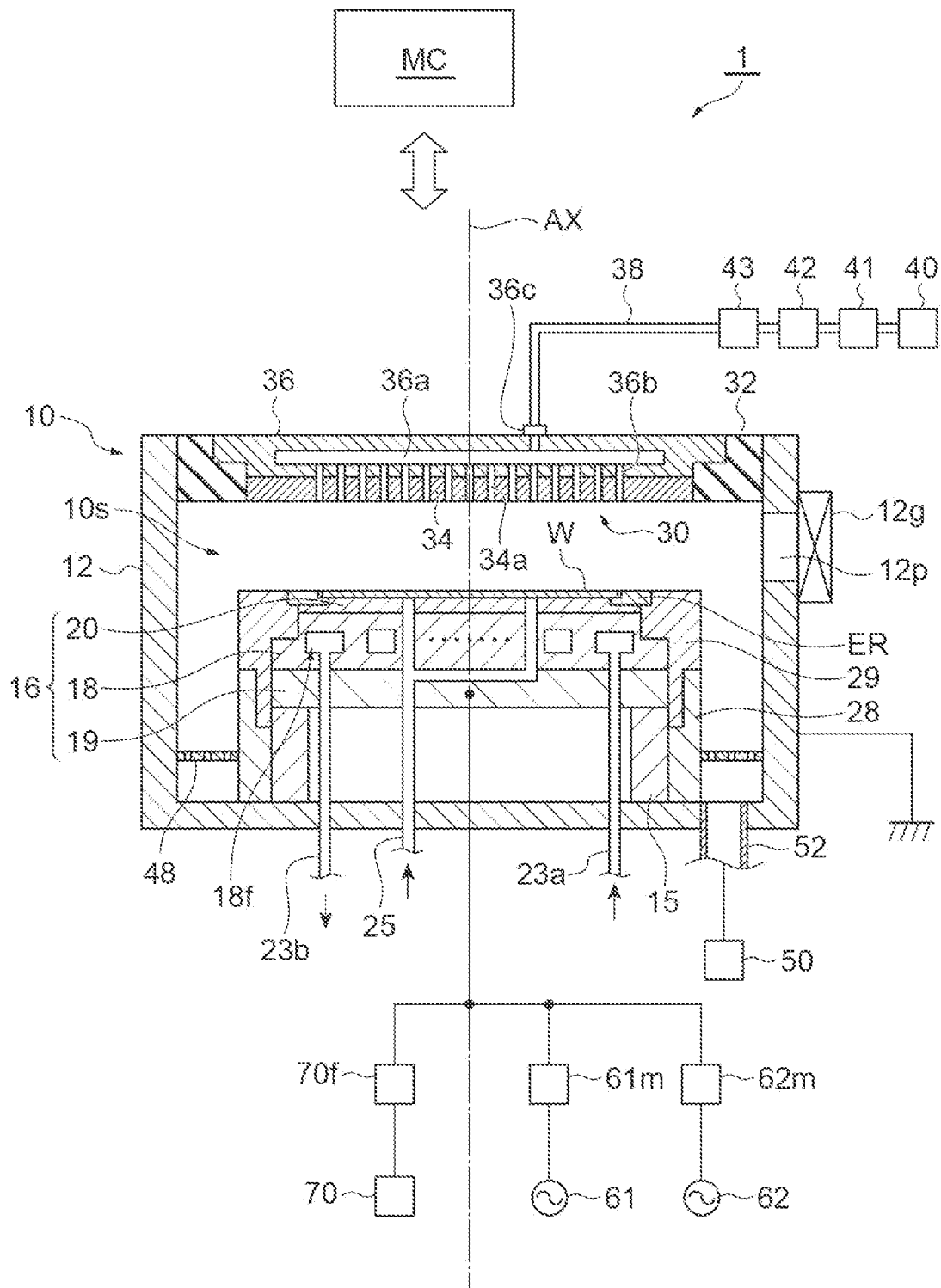


Fig. 2

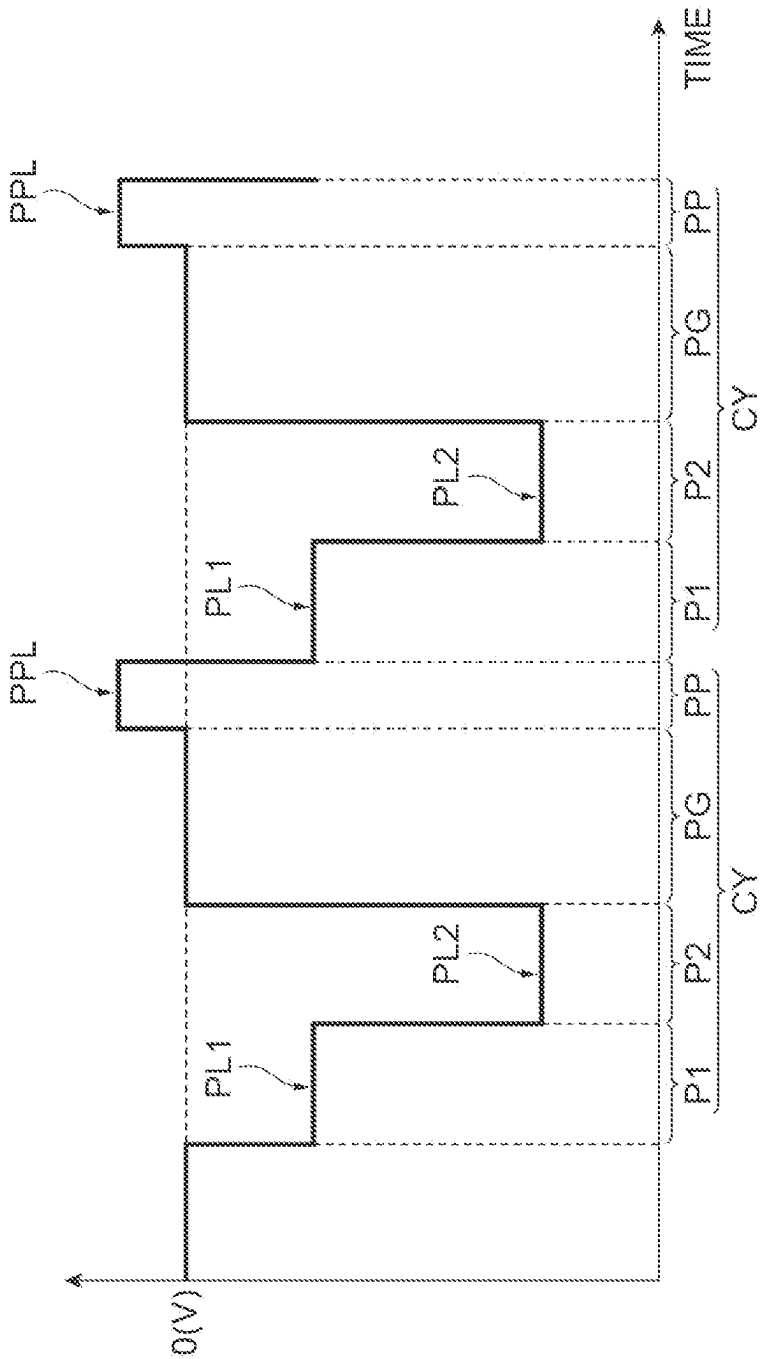


Fig. 3

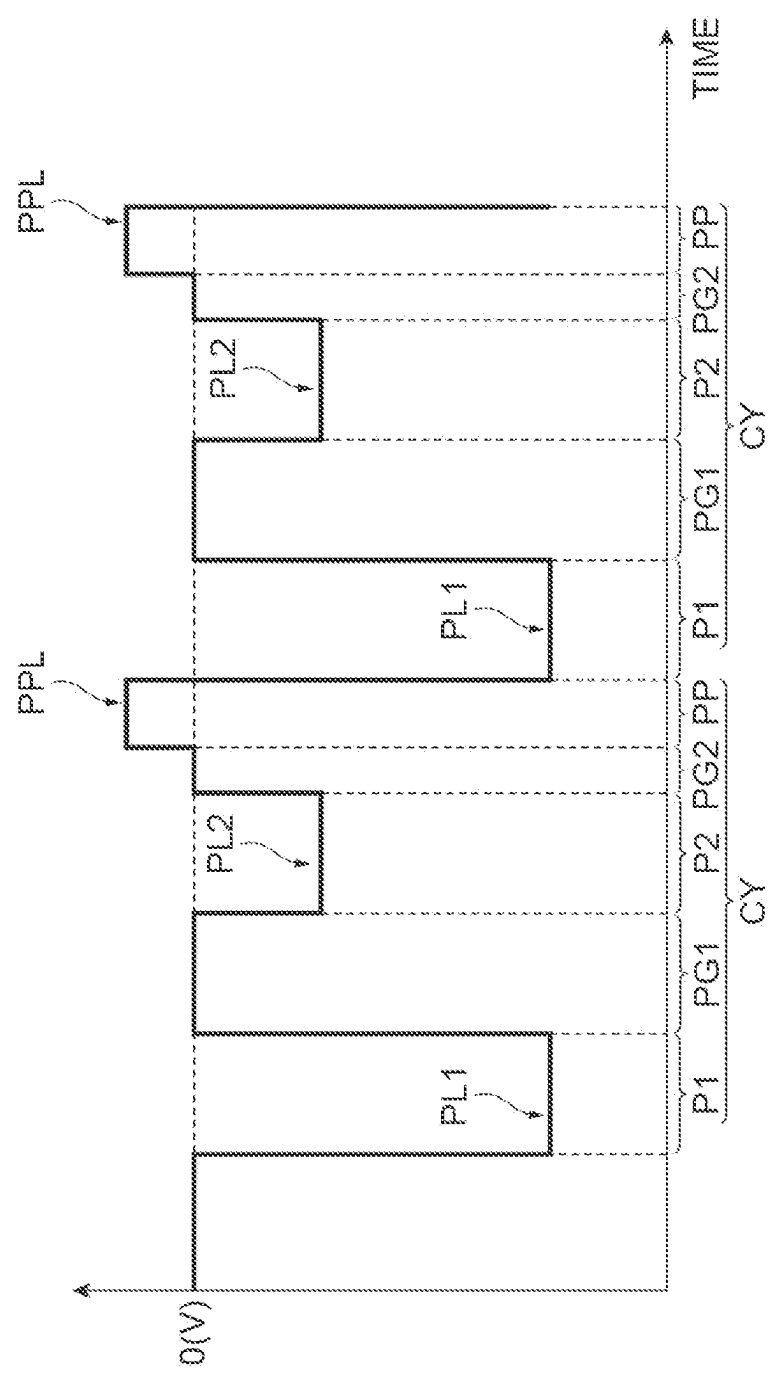


Fig. 4

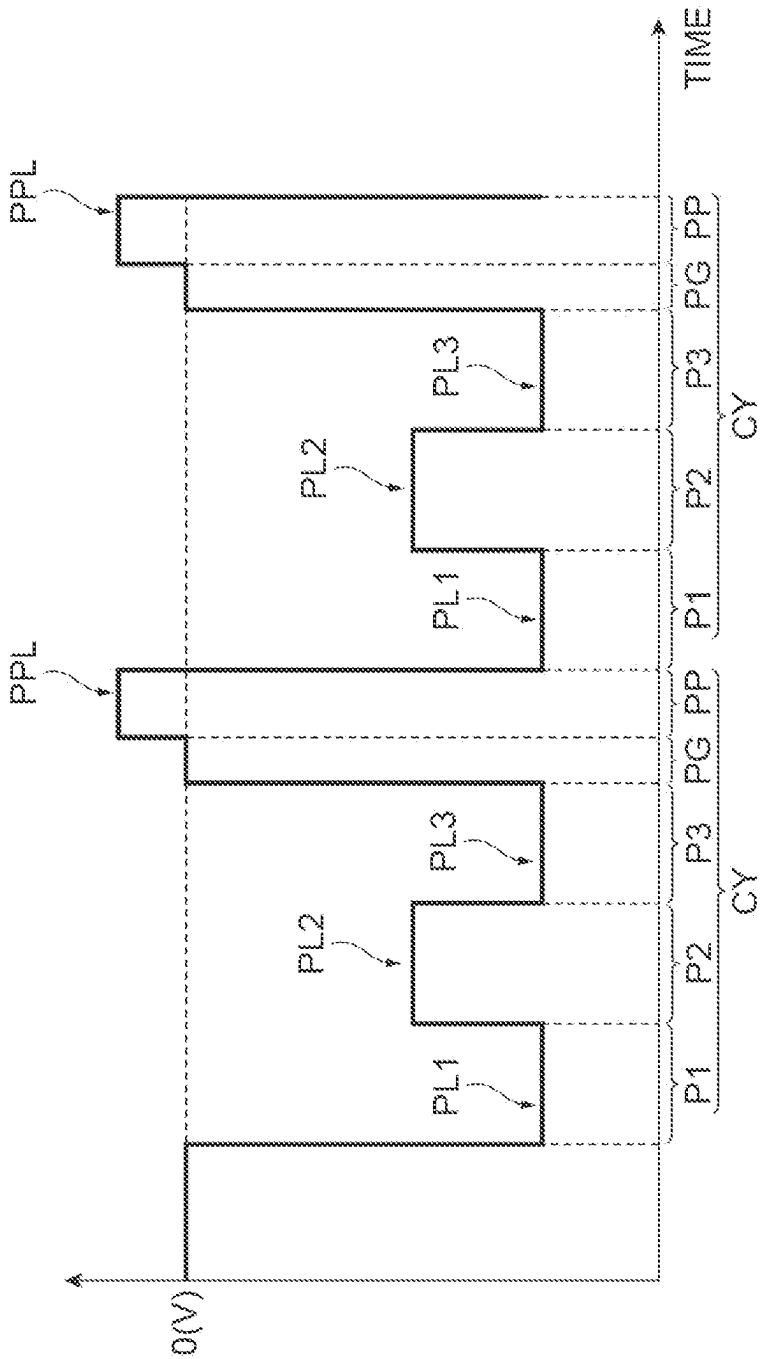


Fig. 5

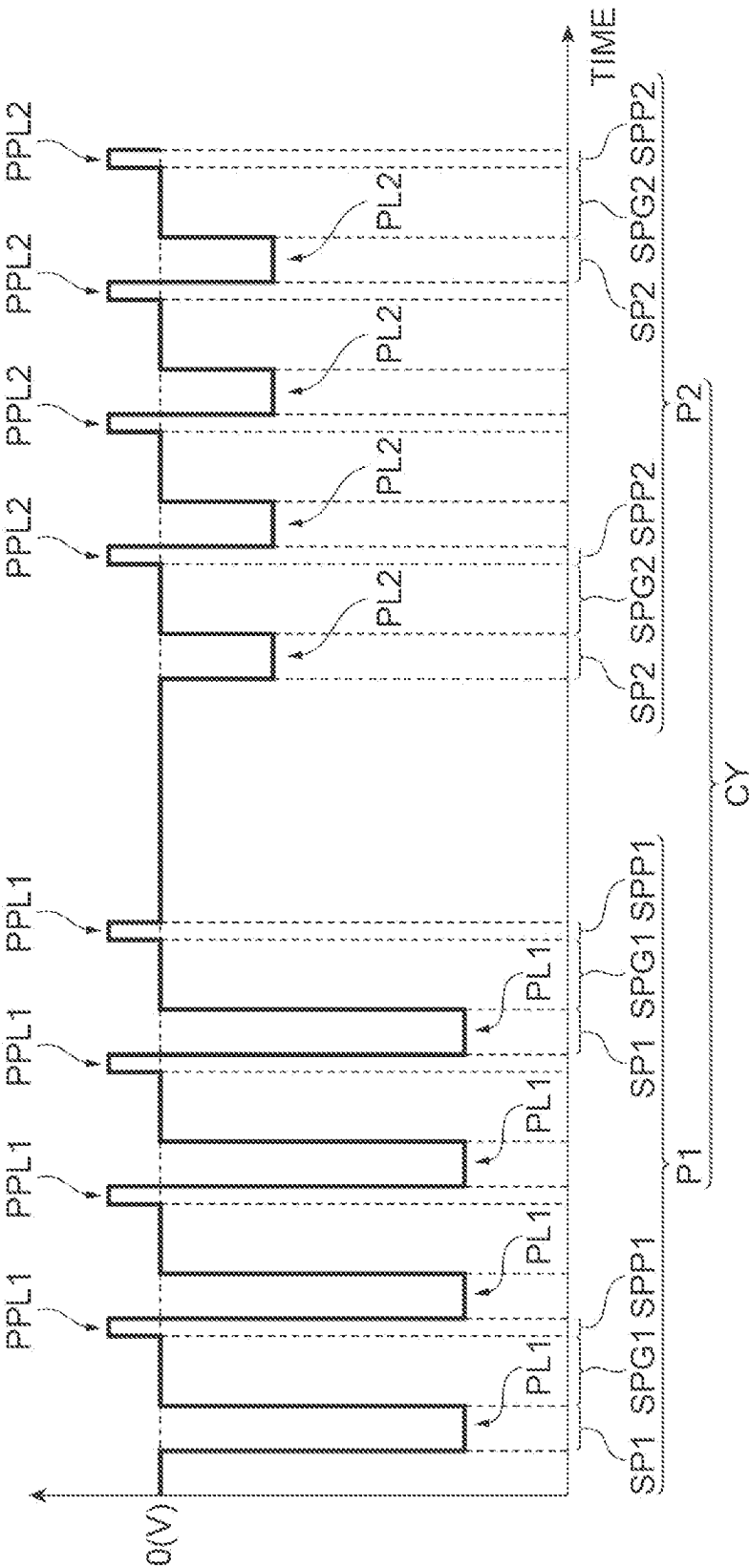


Fig.6

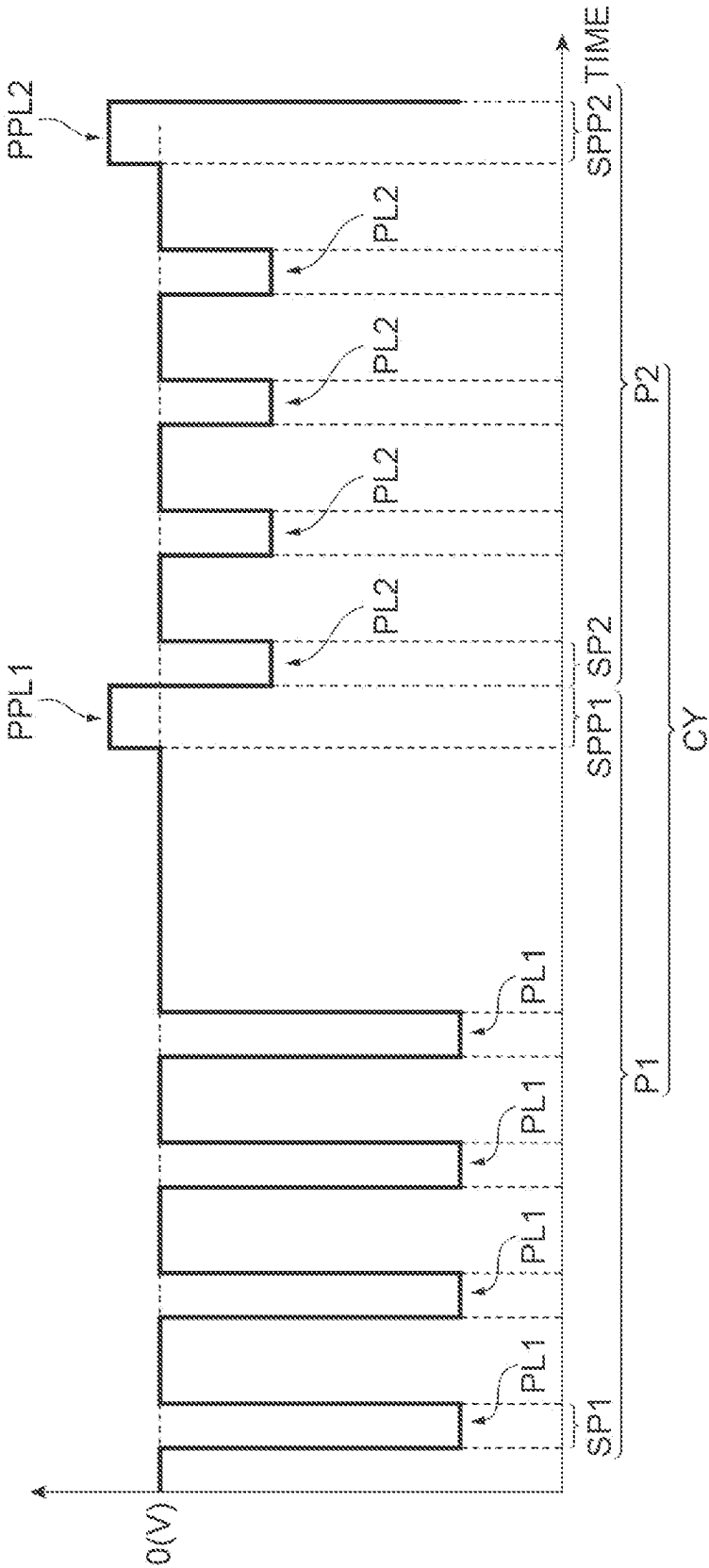


Fig. 7

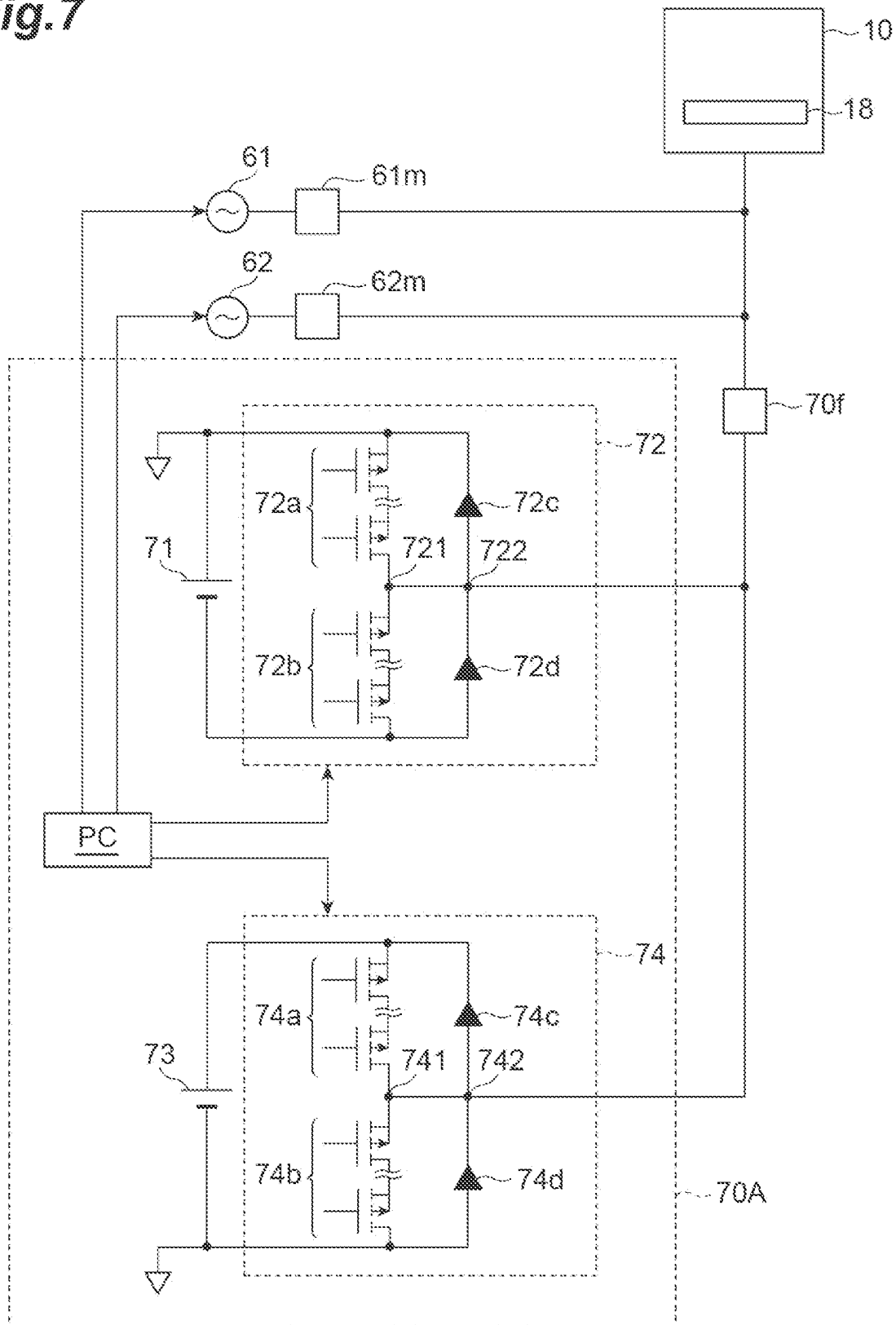
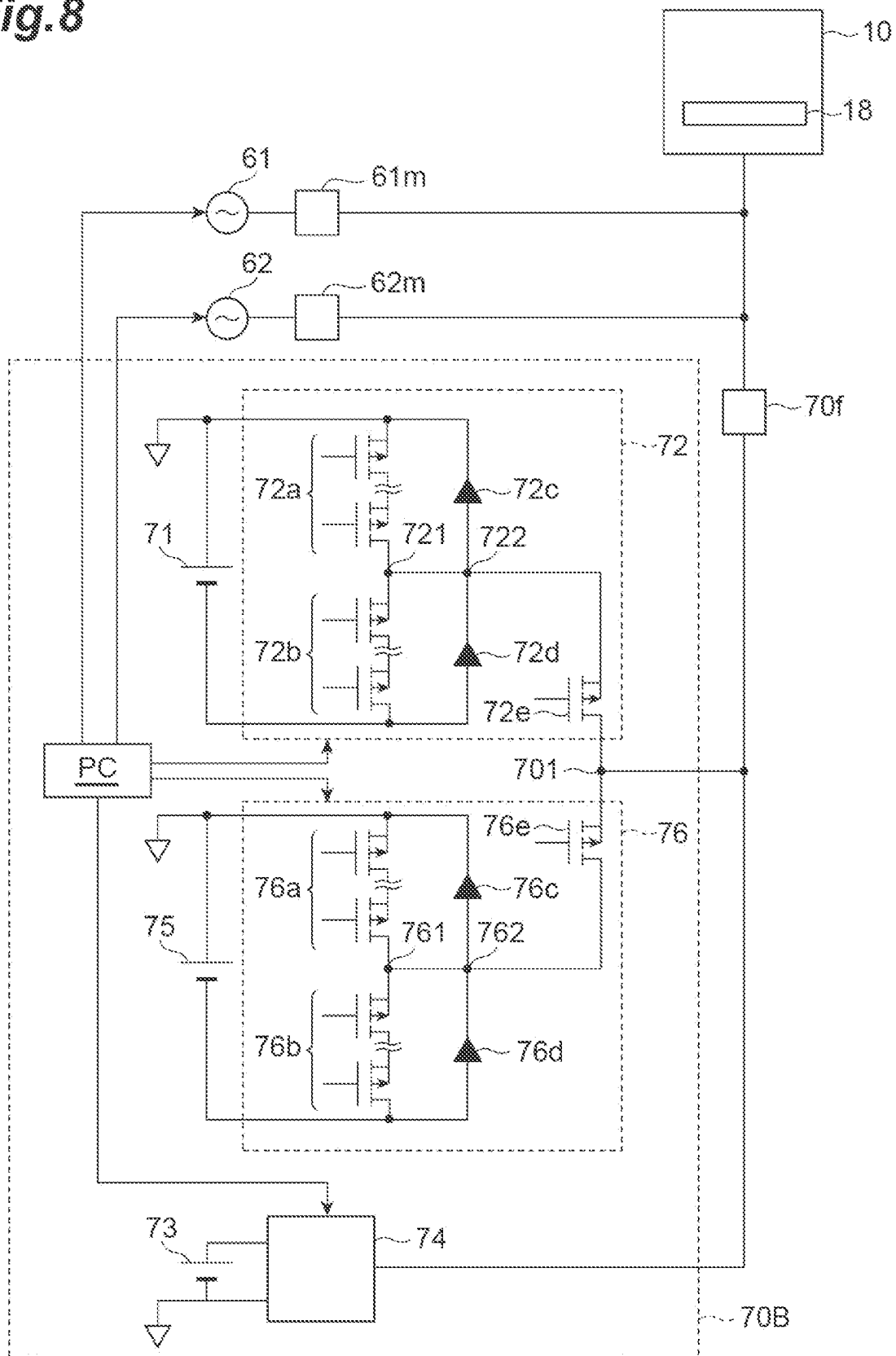


Fig.8



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PLASMA PROCESSING APPARATUS AND PLASMA PROCESSING METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 17/232,231 filed Apr. 16, 2021, which is based on and claims the benefit of priority from Japanese Patent Application Nos. 2021-035192, filed on Mar. 5, 2021 and 2020-079517, filed on Apr. 28, 2020, the entire contents of which are incorporated herein by reference.

FIELD

Exemplary embodiments of the present disclosure relate to a plasma processing apparatus and a plasma processing method.

BACKGROUND

A plasma processing apparatus is used in plasma processing on a substrate. The plasma processing apparatus is provided with a chamber and a substrate holding electrode. The substrate holding electrode is provided in the chamber. The substrate holding electrode holds the substrate placed on the principal surface thereof. A type of such plasma processing apparatus is disclosed in Japanese Unexamined Patent Publication No. 2009-187975.

The plasma processing apparatus disclosed in Japanese Unexamined Patent Publication No. 2009-187975 is further provided with a radio frequency generation device and a DC negative pulse generation device. The radio frequency generation device applies a radio frequency voltage to the substrate holding electrode. In the plasma processing apparatus disclosed in Japanese Unexamined Patent Publication No. 2009-187975, the radio frequency voltage is switched on and off alternately. Further, in the plasma processing apparatus disclosed in Japanese Unexamined Patent Publication No. 2009-187975, a DC negative pulse voltage is applied from the DC negative pulse generation device to the substrate holding electrode according to an on/off timing of the radio frequency voltage.

SUMMARY

In an exemplary embodiment, a plasma processing apparatus is provided. The plasma processing apparatus is provided with a chamber, a substrate support, and a power source system. The substrate support has an electrode and configured to support a substrate in the chamber. The power source system is electrically connected to the electrode and configured to apply a bias voltage to the electrode to draw ions from a plasma in the chamber into the substrate on the substrate support. The power source system is configured to output a first pulse to the electrode in a first period and output a second pulse to the electrode in a second period after the first period, as the bias voltage. Each of the first pulse and the second pulse is a pulse of a voltage. A voltage level of the first pulse is different from a voltage level of the second pulse.

The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, exemplary embodiments, and features described above, further aspects, exemplary embodiments,

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and features will become apparent by reference to the drawings and the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically illustrates a plasma processing apparatus according to an exemplary embodiment.

FIG. 2 is a timing chart of a bias voltage of a first example.

FIG. 3 is a timing chart of a bias voltage of a second example.

FIG. 4 is a timing chart of a bias voltage of a third example.

FIG. 5 is a timing chart of a bias voltage of a fourth example.

FIG. 6 is a timing chart of a bias voltage of a fifth example.

FIG. 7 illustrates a power source system according to an exemplary embodiment.

FIG. 8 illustrates a power source system according to another exemplary embodiment.

FIG. 9 illustrates a configuration of a substrate support according to another exemplary embodiment.

DETAILED DESCRIPTION

Hereinafter, various exemplary embodiments will be described.

In an exemplary embodiment, a plasma processing apparatus is provided. The plasma processing apparatus is provided with a chamber, a substrate support, and a power source system. The substrate support has an electrode and configured to support a substrate in the chamber. The power source system is electrically connected to the electrode and configured to apply a bias voltage to the electrode to draw ions from a plasma in the chamber into the substrate on the substrate support. The power source system is configured to output a first pulse to the electrode in a first period and output a second pulse to the electrode in a second period after the first period, as the bias voltage. Each of the first pulse and the second pulse is a pulse of a voltage. A voltage level of the first pulse is different from a voltage level of the second pulse.

In the aforementioned embodiment, the voltage level of the first pulse is different from the voltage level of the second pulse. Therefore, energy of ions supplied from the plasma to the substrate in the first period is different from energy of ions supplied from the plasma to the substrate in the second period. Hence, according to the aforementioned embodiment, it is possible to supply ions having different energies to the substrate.

In an exemplary embodiment, the second period may be continuous with the first period. Each of the first pulse and the second pulse may be a pulse of a negative voltage. An absolute value of the voltage level of the first pulse may be smaller than an absolute value of the voltage level of the second pulse.

In an exemplary embodiment, each of the first pulse and the second pulse may be a pulse of a negative voltage. An absolute value of the voltage level of the first pulse may be larger than an absolute value of the voltage level of the second pulse. The power source system may be configured such that an output voltage to the electrode is 0 V in a period between the first period and the second period.

In an exemplary embodiment, the second period may be continuous with the first period. Each of the first pulse and the second pulse may be a pulse of a negative voltage. An

absolute value of the voltage level of the first pulse may be larger than an absolute value of the voltage level of the second pulse.

In an exemplary embodiment, the power source system may be configured to output a third pulse to the electrode in a third period after the second period. The third pulse may be a pulse of a negative voltage. An absolute value of a voltage level of the third pulse may be larger than the absolute value of the voltage level of the second pulse.

In an exemplary embodiment, each of the first pulse and the second pulse may be a pulse of a negative voltage. The power source system may be configured to output a pulse of a positive voltage to the electrode before start of the first period within a subsequent cycle of two cycles, each of which includes the first period and the second period. According to the embodiment, electrons are supplied to the substrate when the pulse of the positive voltage is supplied to the electrode of the substrate support. As a result, the amount of positive charges of the substrate is reduced.

In an exemplary embodiment, the power source system may be configured to intermittently output the first pulse to the electrode in the first period. The power source system may be configured to intermittently output the second pulse to the electrode in the second period.

In an exemplary embodiment, each of the first pulse and the second pulse may be a pulse of a negative voltage. The power source system may be configured to alternately output the first pulse and a pulse of a positive voltage to the electrode in the first period. The power source system may be configured to alternately output the second pulse and a pulse of a positive voltage to the electrode in the second period. According to the embodiment, electrons are supplied to the substrate when the pulse of the positive voltage is supplied to the electrode of the substrate support. As a result, the amount of positive charges of the substrate is reduced.

In an exemplary embodiment, the power source system may be configured to output a pulse of a positive voltage to the electrode in a period between the first period and the second period. According to the embodiment, electrons are supplied to the substrate when the pulse of the positive voltage is supplied to the electrode of the substrate support. As a result, the amount of positive charges of the substrate is reduced.

In another exemplary embodiment, a plasma processing method is provided. The plasma processing method includes preparing a substrate on a substrate support provided in a chamber of a plasma processing apparatus. The substrate support further includes an electrode. The plasma processing method includes outputting a first pulse from a power source system to the electrode in a first period as a bias voltage to draw ions from plasma in the chamber into the substrate. The plasma processing method further includes outputting a second pulse as the bias voltage from the power source system to the electrode in a second period. Each of the first pulse and the second pulse is a pulse of a voltage. A voltage level of the first pulse is different from a voltage level of the second pulse.

In an exemplary embodiment, the second period may be continuous with the first period. Each of the first pulse and the second pulse may be a pulse of a negative voltage. An absolute value of the voltage level of the first pulse may be smaller than an absolute value of the voltage level of the second pulse.

In an exemplary embodiment, each of the first pulse and the second pulse may be a pulse of a negative voltage. An absolute value of the voltage level of the first pulse may be larger than an absolute value of the voltage level of the

second pulse. The plasma processing method may further include setting an output voltage from the power source system to the electrode to 0 V in a period between the first period and the second period.

In an exemplary embodiment, the second period may be continuous with the first period. Each of the first pulse and the second pulse may be a pulse of a negative voltage. An absolute value of the voltage level of the first pulse may be larger than an absolute value of the voltage level of the second pulse.

In an exemplary embodiment, the plasma processing method may further include outputting a third pulse from the power source system to the electrode in a third period after the second period. The third pulse may be a pulse of a negative voltage. An absolute value of a voltage level of the third pulse may be larger than the absolute value of the voltage level of the second pulse.

In an exemplary embodiment, the plasma processing method may further include outputting a pulse of a positive voltage from the power source system to the electrode before start of the first period within a subsequent cycle of two cycles, each of which includes the first period and the second period.

In an exemplary embodiment, the first pulse may be intermittently output from the power source system to the electrode in the first period. The second pulse may be intermittently output from the power source system to the electrode in the second period.

In an exemplary embodiment, each of the first pulse and the second pulse may be a pulse of a negative voltage. The plasma processing method may further include intermittently outputting a pulse of a positive voltage from the power source system to the electrode in the first period. The pulse of the positive voltage may be output alternately with the first pulse. The plasma processing method may further include intermittently outputting a pulse of a positive voltage from the power source system to the electrode in the second period. The pulse of the positive voltage may be output alternately with the second pulse.

In an exemplary embodiment, the plasma processing method may further include outputting a pulse of a positive voltage from the power source system to the electrode in a period between the first period and the second period.

Hereinafter, various exemplary embodiments will be described in detail with reference to the drawings. In the drawings, the same or equivalent portions are denoted by the same reference symbols.

FIG. 1 is a diagram schematically showing a plasma processing apparatus according to an exemplary embodiment. A plasma processing apparatus 1 shown in FIG. 1 is a capacitively coupled plasma processing apparatus. The plasma processing apparatus 1 is provided with a chamber 10. The chamber 10 provides an internal space 10s therein. The central axis of the chamber 10 is an axis AX and extends in a vertical direction.

In an embodiment, the chamber 10 may include a chamber body 12. The chamber body 12 has a substantially cylindrical shape. The internal space 10s is provided in the chamber body 12. The chamber body 12 is formed of, for example, aluminum. The chamber body 12 is electrically grounded. A film having plasma resistance is formed on the inner wall surface of the chamber body 12, that is, a wall surface defining the internal space 10s. This film may be a film formed by anodization or a ceramic film such as a film formed of yttrium oxide.

A passage 12p is formed in a side wall of the chamber body 12. A substrate W passes through the passage 12p when

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it is transferred between the internal space 10s and the outside of the chamber 10. A gate valve 12g is provided along the side wall of the chamber body 12 for opening and closing of the passage 12p.

The plasma processing apparatus 1 is further provided with a substrate support 16. The substrate support 16 is configured to support the substrate W placed thereon in the chamber 10. The substrate W has a substantially disk shape. The substrate support 16 is supported by a support 15. The support 15 extends upward from a bottom portion of the chamber body 12. The support 15 has a substantially cylindrical shape. The support 15 is formed of an insulating material such as quartz.

The substrate support 16 has a lower electrode 18. The substrate support 16 may further have an electrostatic chuck 20. The substrate support 16 may further have an electrode plate 19. The electrode plate 19 is formed of a conductive material such as aluminum and has a substantially disk shape. The lower electrode 18 is provided on the electrode plate 19. The lower electrode 18 is formed of a conductive material such as aluminum and has a substantially disk shape. The lower electrode 18 is electrically connected to the electrode plate 19. The central axes of the lower electrode 18 and the electrode plate 19 substantially coincide with the axis AX.

A flow path 18f is provided in the lower electrode 18. The flow path 18f is a flow path for a heat exchange medium. As the heat exchange medium, for example, a refrigerant is used. A circulation device for the heat exchange medium (for example, a chiller unit) is connected to the flow path 18f. The circulation device is provided outside the chamber 10. The heat exchange medium from the circulation device is supplied to the flow path 18f through a pipe 23a. The heat exchange medium supplied to the flow path 18f is returned to the circulation device through a pipe 23b.

The electrostatic chuck 20 is provided on the lower electrode 18. When the substrate W is processed in the internal space 10s, the substrate W is placed on the electrostatic chuck 20 such that the center thereof is located on the axis AX. The electrostatic chuck 20 is configured to hold the substrate. The electrostatic chuck 20 has a main body and an electrode. The main body of the electrostatic chuck 20 is formed of a dielectric such as aluminum oxide or aluminum nitride. The main body of the electrostatic chuck 20 has a substantially disk shape. The central axis of the electrostatic chuck 20 substantially coincides with the axis AX.

The electrode of the electrostatic chuck 20 is provided in the main body of the electrostatic chuck 20. The electrode of the electrostatic chuck 20 has a film formed of a conductor. A direct-current power source is electrically connected to the electrode of the electrostatic chuck 20. When a direct-current voltage is applied from the direct-current power source to the electrode of the electrostatic chuck 20, an electrostatic attraction force is generated between the electrostatic chuck 20 and the substrate W. Due to the generated electrostatic attraction force, the substrate W is attracted to the electrostatic chuck 20 and held by the electrostatic chuck 20.

The substrate support 16 may further support an edge ring ER that is mounted thereon. The edge ring ER has a ring shape and is formed of, for example, silicon or silicon carbide. The edge ring ER is mounted on the substrate support 16 such that the central axis thereof is located on the axis AX. In an embodiment, the edge ring ER may be partially mounted on the electrostatic chuck 20. The substrate W is disposed on the electrostatic chuck 20 and in a region surrounded by the edge ring ER.

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The plasma processing apparatus 1 may be further provided with a gas supply line 25. The gas supply line 25 supplies a heat transfer gas, for example, a He gas, from a gas supply mechanism to a gap between the upper surface of the electrostatic chuck 20 and the rear surface (lower surface) of the substrate W.

The plasma processing apparatus 1 may be further provided with a tubular part 28 and an insulating part 29. The tubular part 28 extends upward from the bottom portion of the chamber body 12. The tubular part 28 extends along the outer periphery of the support 15. The tubular part 28 is formed of a conductive material and has a substantially cylindrical shape. The tubular part 28 is electrically grounded. The insulating part 29 is provided on the tubular part 28. The insulating part 29 is formed of an insulating material. The insulating part 29 is formed of ceramic such as quartz, for example. The insulating part 29 has a substantially cylindrical shape. The insulating part 29 extends along the outer periphery of the electrode plate 19, the outer periphery of the lower electrode 18, and the outer periphery of the electrostatic chuck 20.

The plasma processing apparatus 1 is further provided with an upper electrode 30. The upper electrode 30 is provided above the substrate support 16. The upper electrode 30 closes an upper opening of the chamber body 12 together with a member 32. The member 32 is formed of an insulating material. The upper electrode 30 is supported on an upper portion of the chamber body 12 through the member 32.

The upper electrode 30 may include a ceiling plate 34 and a support 36. The lower surface of the ceiling plate 34 defines the internal space 10s. A plurality of gas discharge holes 34a are formed in the ceiling plate 34. The plurality of gas discharge holes 34a penetrate the ceiling plate 34 in a plate thickness direction (the vertical direction). The ceiling plate 34 is formed of, for example, silicon. Alternatively, the ceiling plate 34 may have a structure in which a plasma-resistant film is provided on the surface of a member made of aluminum. This film may be a film formed by anodization or a ceramic film such as a film formed of yttrium oxide.

The support 36 detachably supports the ceiling plate 34. The support 36 is formed of a conductive material such as aluminum. A gas diffusion chamber 36a is provided in the interior of the support 36. A plurality of gas holes 36b extend downward from the gas diffusion chamber 36a. The plurality of gas holes 36b communicate with the plurality of gas discharge holes 34a, respectively. The support 36 has a gas introduction port 36c formed therein. The gas introduction port 36c is connected to the gas diffusion chamber 36a. A gas supply pipe 38 is connected to the gas introduction port 36c.

A gas source group 40 is connected to the gas supply pipe 38 through a valve group 41, a flow rate controller group 42, and a valve group 43. The gas source group 40, the valve group 41, the flow rate controller group 42, and the valve group 43 configure a gas supply unit. The gas source group 40 includes a plurality of gas sources. Each of the valve group 41 and the valve group 43 includes a plurality of valves (for example, on-off valves). The flow rate controller group 42 includes a plurality of flow rate controllers. Each of the plurality of flow rate controllers of the flow rate controller group 42 is a mass flow controller or a pressure control type flow rate controller. Each of the plurality of gas sources of the gas source group 40 is connected to the gas supply pipe 38 through a corresponding valve of the valve group 41, a corresponding flow rate controller of the flow rate controller group 42, and a corresponding valve of the valve group 43. The plasma processing apparatus 1 can

supply gases from one or more gas sources selected from the plurality of gas sources of the gas source group 40 to the internal space 10s at individually adjusted flow rates.

A baffle member 48 is provided between the tubular part 28 and the side wall of the chamber body 12. The baffle member 48 may be a plate-shaped member. The baffle member 48 may be configured, for example, by coating a plate member made of aluminum with ceramic such as yttrium oxide. A plurality of through-holes are formed in the baffle member 48. An exhaust pipe 52 is connected to the bottom portion of the chamber body 12 below the baffle member 48. An exhaust device 50 is connected to the exhaust pipe 52. The exhaust device 50 has a pressure controller such as an automatic pressure control valve, and a vacuum pump such as a turbo molecular pump, and is capable of reducing the pressure in the internal space 10s.

The plasma processing apparatus 1 is further provided with a radio frequency power source 61. The radio frequency power source 61 is a power source that generates first radio frequency power for plasma generation. The frequency of the first radio frequency power is a frequency within the range of 27 to 100 MHz, for example, a frequency of 40 MHz or 60 MHz. The radio frequency power source 61 is connected to the lower electrode 18 through a matcher 61m and the electrode plate 19. The matcher 61m has a matching circuit for matching the impedance on the load side (the lower electrode 18 side) of the radio frequency power source 61 with the output impedance of the radio frequency power source 61. The radio frequency power source 61 does not need to be electrically connected to the lower electrode 18, and may be connected to the upper electrode 30 through the matcher 61m.

The plasma processing apparatus 1 may be further provided with a radio frequency power source 62. The radio frequency power source 62 is a power source that generates second radio frequency power. The frequency of the second radio frequency power is lower than the frequency of the first radio frequency power. The radio frequency power source 62 is connected to the lower electrode 18 through a matcher 62m and the electrode plate 19. The matcher 62m has a matching circuit for matching the impedance on the load side (the lower electrode 18 side) of the radio frequency power source 62 with the output impedance of the radio frequency power source 62. The plasma processing apparatus 1 does not need to include the radio frequency power source 62 and the matcher 62m.

In the plasma processing apparatus 1, a gas is supplied from the gas supply unit to the internal space 10s. Then, the radio frequency power is supplied, whereby the gas is excited in the internal space 10s. As a result, plasma is generated in the internal space 10s. The substrate W is processed with chemical species such as ions and/or radicals from the plasma.

The plasma processing apparatus 1 is further provided with a power source system 70. The power source system 70 is electrically connected to the lower electrode 18. The power source system 70 is configured to apply a bias voltage to the lower electrode 18 to draw ions from the plasma into the substrate on the substrate support 16. The power source system 70 may be connected to the lower electrode 18 through a filter 70f. The filter 70f includes a filter circuit that cuts off or reduces the radio frequency power toward the power source system 70. The details of the power source system 70 will be described later.

The plasma processing apparatus 1 may be further provided with a controller MC. The controller MC is a computer which includes a processor, a storage device, an input

device, a display device, and the like, and controls each part of the plasma processing apparatus 1. Specifically, the controller MC executes a control program stored in the storage device and controls each part of the plasma processing apparatus 1, based on recipe data stored in the storage device. A process designated by the recipe data is executed in the plasma processing apparatus 1 under the control by the controller MC. Plasma processing methods according to various embodiments may be performed in the plasma processing apparatus 1 under the control of each part of the plasma processing apparatus 1 by the controller MC.

Hereinafter, the bias voltage that is generated by the power source system 70 will be described with reference to FIGS. 2 to 6. FIGS. 2 to 6 are timing charts of the bias voltages of first to fifth examples, respectively. Further, the plasma processing methods according to various exemplary embodiments will also be described below.

In various exemplary embodiments, the plasma processing method includes a step of preparing the substrate W on the substrate support 16. The plasma processing method is performed in a state where the substrate W is placed on the substrate support 16.

The plasma processing method includes a step of generating plasma in the chamber 10. A processing gas is supplied from the gas supply unit into the chamber 10 in order to generate plasma. Further, the pressure of the gas in the chamber 10 is adjusted by the exhaust device 50 to a designated pressure. Further, the first radio frequency power is supplied from the radio frequency power source 61. As a result, plasma is generated from the processing gas in the chamber 10. The second radio frequency power from the radio frequency power source 62 may be supplied to the lower electrode 18 during the generation of the plasma. The steps of the plasma processing method that are described below are executed in a period in which the plasma is being generated in the chamber 10.

The plasma processing method includes a step of outputting a first pulse PL1 as the bias voltage from the power source system 70 to the lower electrode 18 in a first period P1, as shown in FIGS. 2 to 6. The first pulse PL1 is a pulse of a voltage. The first pulse PL1 may be a pulse of a direct-current voltage. The first pulse PL1 may have a waveform other than a rectangular wave such as a triangular wave or an impulse wave. The first pulse may have a waveform whose voltage value changes at the leading and trailing edges thereof.

The plasma processing method includes a step of outputting a second pulse PL2 as the bias voltage from the power source system 70 to the lower electrode 18 in a second period P2, as shown in FIGS. 2 to 6. The second period P2 is a period after the first period P1. The second pulse PL2 is a pulse of a voltage. The voltage level of the first pulse PL1 is different from the voltage level of the second pulse PL2. The second pulse PL2 may be a pulse of a direct-current voltage. The second pulse PL2 may have a waveform other than a rectangular wave such as a triangular wave or an impulse wave. The second pulse may have a waveform whose voltage value changes at the leading and trailing edges thereof.

In each of the examples shown in FIGS. 2 to 6, each of the first pulse PL1 and the second pulse PL2 is a pulse of a negative voltage (for example, a negative direct-current voltage). The difference between the voltage level of one (hereinafter referred to as an "H-level pulse") of the first pulse PL1 and the second pulse PL2 and the reference level (for example, 0 V) may be 6 kV or more. The difference between the voltage level of the H-level pulse and the

reference level may be 10 kV or 20 kV or more. The difference between the voltage level of the other (hereinafter referred to as an "L-level pulse") of the first pulse PL1 and the second pulse PL2 and the reference level (for example, 0 V) may be 5 kV or less.

In the plasma processing method, as shown in FIGS. 2 to 6, a cycle CY may be repeated. The cycle CY includes the first period P1 and the second period P2. In the first to third examples shown in FIGS. 2 to 4, the frequency defining the cycle CY may be a frequency of 100 kHz or more and 1 MHz or less. In the fourth and fifth examples shown in FIGS. 5 and 6, the frequency defining the cycle CY may be 0.2 Hz or more and 1 Hz or less.

In an embodiment, as in the first example shown in FIG. 2, the second period P2 may be continuous with the first period P1. In this embodiment, the absolute value of the voltage level of the first pulse PL1 may be smaller than the absolute value of the voltage level of the second pulse PL2.

In the first example shown in FIG. 2, the reference level of the bias voltage is 0 V. This reference level may have a positive or negative value. In the first example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the first pulse PL1 is smaller than the difference between the reference level and the voltage level of the second pulse PL2. That is, in the first example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the voltage level of the second pulse PL2 is lower than the voltage level of the first pulse PL1.

In another embodiment, as in the second example shown in FIG. 3, the absolute value of the voltage level of the first pulse PL1 may be larger than the absolute value of the voltage level of the second pulse PL2. In this embodiment, the plasma processing method may further include a step of setting the output voltage from the power source system 70 to the lower electrode 18 to 0 V in a period PG1 between the first period P1 and the second period P2.

Also in the second example shown in FIG. 3, the reference level of the bias voltage is 0 V. This reference level may have a positive or negative value. In the second example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the first pulse PL1 is larger than the difference between the reference level and the voltage level of the second pulse PL2. That is, in the second example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the voltage level of the first pulse PL1 is lower than the voltage level of the second pulse PL2.

In still another embodiment, as in the third example shown in FIG. 4, the second period P2 may be continuous with the first period P1. In this embodiment, the absolute value of the voltage level of the first pulse PL1 may be larger than the absolute value of the voltage level of the second pulse PL2.

In still another embodiment, as in the third example shown in FIG. 4, the plasma processing method may further include a step of outputting a third pulse PL3 from the power source system 70 to the lower electrode 18 in a third period P3 after the second period P2. The third pulse PL3 may be a pulse of a negative voltage, and the absolute value of the voltage level of the third pulse PL3 may be larger than the absolute value of the voltage level of the second pulse PL2. The absolute value of the voltage level of the third pulse PL3

may be the same as the absolute value of the voltage level of the first pulse PL1. That is, the third pulse PL3 may be an H-level pulse. The difference between the voltage level of the third pulse PL3 and the reference level (for example, 0 V) may be 6 kV or more. The difference between the voltage level of the third pulse PL3 and the reference level may be 10 kV or 20 kV or more.

Also in the third example shown in FIG. 4, the reference level of the bias voltage is 0 V. This reference level may have a positive or negative value. In the third example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the first pulse PL1 is larger than the difference between the reference level and the voltage level of the second pulse PL2. That is, in the third example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the voltage level of the first pulse PL1 is lower than the voltage level of the second pulse PL2. Further, in the third example, each of the voltage level of the third pulse PL3 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the third pulse PL3 is larger than the difference between the reference level and the voltage level of the second pulse PL2. That is, in the third example, each of the voltage level of the third pulse PL3 and the voltage level of the second pulse PL2 may have any value, as long as the voltage level of the third pulse PL3 is lower than the voltage level of the second pulse PL2.

In still another embodiment, the plasma processing method may further include a step of outputting a pulse PPL of a positive voltage, as in the first to third examples shown in FIGS. 2 to 4. The voltage level of the pulse PPL of the voltage is higher than the reference level (for example, 0 V). The pulse PPL of the voltage may be a pulse of a positive direct-current voltage. The pulse PPL is output in a period PP. The period PP is a period before the start of the first period P1 in the subsequent cycle CY of any two consecutive cycles CY. The period PP may be a period immediately before the subsequent first period P1. In this case, it becomes possible to supply ions having high energy to the substrate W in the subsequent first period P1.

In the first example shown in FIG. 2, the period PP is a period after a period PG. In the first example shown in FIG. 2, the period PG is a period after the second period P2. In the first example shown in FIG. 2, in the period PG, the output voltage from the power source system 70 to the lower electrode 18 is set to 0 V (that is, the reference level).

In the second example shown in FIG. 3, the period PP is a period after a period PG2. In the second example shown in FIG. 3, the period PG2 is a period after the second period P2. In the second example shown in FIG. 3, in the period PG2, the output voltage from the power source system 70 to the lower electrode 18 is set to 0 V.

In the third example shown in FIG. 4, the period PP is a period after the period PG. In the third example shown in FIG. 4, the period PG is a period after the third period P3. In the third example shown in FIG. 4, in the period PG, the output voltage from the power source system 70 to the lower electrode 18 is set to 0 V (for example, the reference level).

In still another embodiment, as in the fourth example shown in FIG. 5 and the fifth example shown in FIG. 6, the first pulse PL1 may be intermittently output from the power source system 70 to the lower electrode 18 in the first period P1. The first pulse PL1 may be periodically output from the power source system 70 to the lower electrode 18 in the first

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period P1. The first pulse PL1 is output in sub-periods SP1 within the first period P1. The first pulse PL1 may be periodically output at a frequency of 100 kHz or more and 1 MHz or less.

In still another embodiment, as in the fourth example shown in FIG. 5 and the fifth example shown in FIG. 6, the second pulse PL2 may be intermittently output from the power source system 70 to the lower electrode 18 in the second period P2. The second pulse PL2 may be periodically output from the power source system 70 to the lower electrode 18 in the second period P2. The second pulse PL2 is output in sub-periods SP2 within the second period P2. The second pulse PL2 may be periodically output at a frequency of 100 kHz or more and 1 MHz or less.

In the fourth example shown in FIG. 5 and the fifth example shown in FIG. 6, the absolute value of the voltage level of the first pulse PL1 is larger than the absolute value of the voltage level of the second pulse PL2. The absolute value of the voltage level of the first pulse PL1 may be smaller than the absolute value of the voltage level of the second pulse PL2.

Also in the fourth example shown in FIG. 5, the reference level of the bias voltage is 0 V. This reference level may have a positive or negative value. In the fourth example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the first pulse PL1 is different from the difference between the reference level and the voltage level of the second pulse PL2.

In still another embodiment, the plasma processing method may further include a step of intermittently outputting a pulse PPL1 of a positive voltage from the power source system 70 to the lower electrode 18 in the first period P1, as in the fourth example shown in FIG. 5. The voltage level of the pulse PPL1 of the voltage is higher than the reference level (for example, 0 V). The pulse PPL1 of the voltage may be a pulse of a positive direct-current voltage. In this embodiment, the pulse PPL1 of the positive voltage may be output alternately with the first pulse PL1. The pulse PPL1 of the positive voltage is output in periods SPP1. The periods SPP1 may be periods after the respective periods SPG1. The periods SPG1 are periods after the respective sub-periods SP1. In the periods SPG1, the output voltage from the power source system 70 to the lower electrode 18 may be set to 0 V (that is, the reference level). The periods SPP1 may be periods immediately before the respective subsequent sub-periods SP1. In this case, it becomes possible to supply ions having high energy to the substrate W in the subsequent sub-periods SP1.

In still another embodiment, the plasma processing method may further include a step of intermittently outputting a pulse PPL2 of a positive voltage from the power source system 70 to the lower electrode 18 in the second period P2, as in the fourth example shown in FIG. 5. The voltage level of the pulse PPL2 of the voltage is higher than the reference level (for example, 0 V). The pulse PPL2 of the voltage may be a pulse of a positive direct-current voltage. In this embodiment, the pulse PPL2 of the positive voltage may be output alternately with the second pulse PL2. The pulse PPL2 of the positive voltage is output in periods SPP2. The periods SPP2 may be periods after the respective periods SPG2. The periods SPG2 are periods after the respective sub-periods SP2. In the periods SPG2, the output voltage from the power source system 70 to the lower electrode 18 may be set to 0 V (that is, the reference level). The periods SPP2 may be periods immediately before the

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respective subsequent sub-periods SP2. In this case, it becomes possible to supply ions having high energy to the substrate W in the subsequent sub-period SP2.

In still another embodiment, the plasma processing method may further include a step of outputting the pulse PPL1 of the positive voltage from the power source system 70 to the lower electrode 18 in the period SPP1 between the first period P1 and the second period P2, as in the fifth example shown in FIG. 6. The voltage level of the pulse PPL1 of the voltage is higher than the reference level (for example, 0 V). The pulse PPL1 of the voltage may be a pulse of a positive direct-current voltage. The period SPP1 may be a period immediately before the subsequent sub-period SP2. In this case, it becomes possible to supply ions having high energy to the substrate W in the subsequent sub-period SP2.

In still another embodiment, the plasma processing method may further include a step of outputting the pulse PPL2 of the positive voltage from the power source system 70 to the lower electrode 18 in the period SPP2, as in the fifth example shown in FIG. 6. The voltage level of the pulse PPL2 of the voltage is higher than the reference level (for example, 0 V). The pulse PPL2 of the voltage may be a pulse of a positive direct-current voltage. The period SPP2 may be a period between the second period P2 and the subsequent first period P1. The period SPP2 may be a period immediately before the subsequent sub-period SP1. In this case, it becomes possible to supply ions having high energy to the substrate W in the subsequent sub-period SP1.

Also in the fifth example shown in FIG. 6, the reference level of the bias voltage is 0 V. This reference level may have a positive or negative value. In the fifth example, each of the voltage level of the first pulse PL1 and the voltage level of the second pulse PL2 may have any value, as long as the difference between the reference level and the voltage level of the first pulse PL1 is different from the difference between the reference level and the voltage level of the second pulse PL2.

In the various exemplary embodiments described above, the energy of the ions that are supplied from the plasma to the substrate W in the first period P1 and the energy of the ions that are supplied from the plasma to the substrate W in the second period P2 are different. Therefore, it becomes possible to supply ions having different energies to the substrate W.

Further, in some exemplary embodiments, the pulse of the positive direct-current voltage is supplied to the lower electrode 18. The pulse of the positive direct-current voltage is supplied to the lower electrode 18, whereby electrons are supplied to the substrate W. As a result, the amount of positive charges of the substrate W is reduced.

In a case of forming an opening in the substrate W by etching a film of the substrate W by using the bias voltage of each of the first to third examples, in the period in which the L-level pulse is supplied, deposits on the substrate W are removed, so that the shape of the opening can be adjusted. Further, in this case, ions can be supplied to the bottom of a deep opening in the period in which the H-level pulse is supplied.

The bias voltages of the fourth and fifth examples can be used for etching of a multilayer film having different types of films. The bias voltages of the fourth and fifth examples can be used for etching in the manufacture of a NAND device, for example. In the manufacture of the NAND device, a multilayer film including alternately laminated layers of a silicon oxide film and a silicon nitride film or a polycrystalline silicon film are etched. The silicon oxide film can be etched with ions having relatively high energy, and

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the silicon nitride film or the polycrystalline silicon film can be etched with ions having relatively low energy. In a case of using the bias voltages of the fourth and fifth examples, the silicon oxide film may be etched in the first period P1 and the silicon nitride film or the polycrystalline silicon film may be etched in the second period P2.

Hereinafter, power source systems according to some exemplary embodiments that can be adopted as the power source system 70 of the plasma processing apparatus 1 will be described.

FIG. 7 is a diagram showing a power source system according to an exemplary embodiment. A power source system 70A shown in FIG. 7 can be adopted as the power source system 70 of the plasma processing apparatus 1. The power source system 70A can output the bias voltages of the first to fifth examples described above. The power source system 70A may include a direct-current power source 71, a pulse unit 72, and a pulse controller PC.

The direct-current power source 71 is a power source that generates a negative direct-current voltage. The positive terminal of the direct-current power source 71 is connected to the ground. The direct-current power source 71 may be a variable direct-current power source.

The pulse unit 72 is configured to generate a pulse of a voltage from the negative direct-current voltage from the direct-current power source 71. In an embodiment, the pulse unit 72 may include one or more switching transistors 72a, one or more switching transistors 72b, a diode 72c, and a diode 72d.

The one or more switching transistors 72a are connected between the positive terminal of the direct-current power source 71 and a node 721. In a case where the pulse unit 72 includes a plurality of switching transistors 72a, these switching transistors 72a are connected in series between the positive terminal of the direct-current power source 71 and the node 721.

The one or more switching transistors 72b are connected between the negative terminal of the direct-current power source 71 and the node 721. In a case where the pulse unit 72 includes a plurality of switching transistors 72b, these switching transistors 72b are connected in series between the negative terminal of the direct-current power source 71 and the node 721.

The diode 72c is connected in parallel with the one or more switching transistors 72a. The diode 72c is connected between the positive terminal of the direct-current power source 71 and a node 722. The cathode of the diode 72c is connected to the positive terminal of the direct-current power source 71, and the anode of the diode 72c is connected to the node 722. The node 722 is connected to the node 721.

The diode 72d is connected in parallel with the one or more switching transistors 72b. The diode 72d is connected between the negative terminal of the direct-current power source 71 and the node 722. The anode of the diode 72d is connected to the negative terminal of the direct-current power source 71, and the cathode of the diode 72d is connected to the node 722.

The pulse controller PC is configured to set an output period of the pulse of the voltage from the pulse unit 72 by supplying a control signal to the pulse unit 72. In an embodiment, the pulse controller PC supplies a pulse control signal as the control signal to the control terminals of the one or more switching transistors 72a and the control terminals of the one or more switching transistors 72b.

When the one or more switching transistors 72a enter a closed state and the one or more switching transistors 72b

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enter an open state by the pulse control signal from the pulse controller PC, the node 722 is connected to the ground. As a result, the output voltage of the power source system 70A becomes 0 V (that is, the reference level).

When the one or more switching transistors 72a enter an open state and the one or more switching transistors 72b enter a closed state by the pulse control signal from the pulse controller PC, the node 722 is connected to the negative terminal of the direct-current power source 71. As a result, a pulse of a negative voltage (for example, a negative direct-current voltage) is output from the power source system 70A. Each of the first pulse, the second pulse, and the third pulse described above is generated by adjusting the voltage level of the pulse that is output from the node 722 by adjusting the output voltage level of the direct-current power source 71.

In an embodiment, the power source system 70A may further include a direct-current power source 73 and a pulse unit 74. The direct-current power source 73 is a power source that generates a positive direct-current voltage. The negative terminal of the direct-current power source 73 is connected to the ground. The direct-current power source 73 may be a variable direct-current power source.

The pulse unit 74 is configured to generate a pulse of a voltage from the positive direct-current voltage from the direct-current power source 73. In an embodiment, the pulse unit 74 may include one or more switching transistors 74a, one or more switching transistors 74b, a diode 74c, and a diode 74d.

The one or more switching transistors 74a are connected between the positive terminal of the direct-current power source 73 and a node 741. In a case where the pulse unit 74 includes a plurality of switching transistors 74a, these switching transistors 74a are connected in series between the positive terminal of the direct-current power source 73 and the node 741.

The one or more switching transistors 74b are connected between the negative terminal of the direct-current power source 73 and the node 741. In a case where the pulse unit 74 includes a plurality of switching transistors 74b, these switching transistors 74b are connected in series between the negative terminal of the direct-current power source 73 and the node 741.

The diode 74c is connected in parallel with the one or more switching transistors 74a. The diode 74c is connected between the positive terminal of the direct-current power source 73 and a node 742. The cathode of the diode 74c is connected to the positive terminal of the direct-current power source 73, and the anode of the diode 74c is connected to the node 742. The node 742 is connected to the node 741.

The diode 74d is connected in parallel with the one or more switching transistors 74b. The diode 74d is connected between the negative terminal of the direct-current power source 73 and the node 742. The anode of the diode 74d is connected to the negative terminal of the direct-current power source 73, and the cathode of the diode 74d is connected to the node 742.

The pulse controller PC is configured to set an output period of the pulse of the voltage from the pulse unit 74 by supplying a control signal to the pulse unit 74. In an embodiment, the pulse controller PC supplies a pulse control signal as the control signal to the control terminals of the one or more switching transistors 74a and the control terminals of the one or more switching transistors 74b.

When the one or more switching transistors 74a enter an open state and the one or more switching transistors 74b

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enter a closed state by the pulse control signal from the pulse controller PC, the node 742 is connected to the ground. As a result, the output voltage of the power source system 70A becomes 0 V (that is, the reference level).

When the one or more switching transistors 74a enter a closed state and the one or more switching transistors 74b enter an open state by the pulse control signal from the pulse controller PC, the node 742 is connected to the positive terminal of the direct-current power source 73. As a result, a pulse of a positive voltage (for example, a positive direct-current voltage) is output from the power source system 70A.

When the pulse of the negative voltage is output from the power source system 70A, the one or more switching transistors 74a and the one or more switching transistors 74b may be set to an open state. Further, when the pulse of the positive voltage is output from the power source system 70A, the one or more switching transistors 72a and the one or more switching transistors 72b may be set to an open state.

In an embodiment, the pulse controller PC may supply the pulse control signal to the radio frequency power source 61 and/or the radio frequency power source 62. The radio frequency power source 61 may output the pulse of the first radio frequency power according to the pulse control signal. The pulse of the first radio frequency power may be output in the same phase as, or may be output in a different phase from, the pulse of the direct-current voltage that is output from the power source system 70A. The radio frequency power source 62 may output the pulse of the second radio frequency power according to the pulse control signal. The pulse of the second radio frequency power may be output in the same phase as, or may be output in a different phase from, the pulse of the direct-current voltage that is output from the power source system 70A. The radio frequency power source 61 may output a continuous wave of the first radio frequency power. Further, the radio frequency power source 62 may output a continuous wave of the second radio frequency power.

Hereinafter, FIG. 8 will be referred to. FIG. 8 is a diagram showing a power source system according to another exemplary embodiment. A power source system 70B shown in FIG. 8 can be adopted as the power source system 70 of the plasma processing apparatus 1. The power source system 70B can output the bias voltages of the first to fifth examples described above. Hereinafter, the differences between the power source system 70B and the power source system 70A will be described.

The pulse unit 72 of the power source system 70B further includes a switching transistor 72e. The switching transistor 72e is connected between the node 722 and a node 701. The node 701 is connected to the lower electrode 18.

The switching transistor 72e switches between an open state and a closed state according to the control signal (pulse control signal) that is supplied from the pulse controller PC to the control terminal thereof. When the switching transistor 72e is in an open state, the pulse unit 72 is electrically separated from the node 701 and the lower electrode 18. When the switching transistor 72e is in a closed state, the pulse unit 72 is connected to the node 701 and the lower electrode 18, and a pulse of a negative voltage is output from the pulse unit 72 to the lower electrode 18. The pulse of the negative voltage that is output from the pulse unit 72 to the lower electrode 18 is used as one of the first pulse PL1 and the second pulse PL2. In a case where the pulse unit 72 outputs the first pulse PL1, the pulse unit 72 may further output the third pulse PL3.

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The power source system 70B further includes a direct-current power source 75 and a pulse unit 76. The direct-current power source 75 is a power source that generates a negative direct-current voltage. The positive terminal of the direct-current power source 75 is connected to the ground. The direct-current power source 75 may be a variable direct-current power source. The level of the negative direct-current voltage that is generated by the direct-current power source 75 is different from the level of the negative direct-current voltage that is generated by the direct-current power source 71.

The pulse unit 76 is configured to generate a pulse of a voltage from the negative direct-current voltage from the direct-current power source 75. In an embodiment, the pulse unit 76 may include one or more switching transistors 76a, one or more switching transistors 76b, a diode 76c, a diode 76d, and a switching transistors 76e.

The one or more switching transistors 76a are connected between the positive terminal of the direct-current power source 75 and a node 761. In a case where the pulse unit 76 includes a plurality of switching transistors 76a, these switching transistors 76a are connected in series between the positive terminal of the direct-current power source 75 and the node 761.

The one or more switching transistors 76b are connected between the negative terminal of the direct-current power source 75 and the node 761. In a case where the pulse unit 76 includes a plurality of switching transistors 76b, these switching transistors 76b are connected in series between the negative terminal of the direct-current power source 75 and the node 761.

The diode 76c is connected in parallel with the one or more switching transistors 76a. The diode 76c is connected between the positive terminal of the direct-current power source 75 and a node 762. The cathode of the diode 76c is connected to the positive terminal of the direct-current power source 75, and the anode of the diode 76c is connected to the node 762. The node 762 is connected to the node 761. The node 762 is connected to the node 701 through the switching transistor 76e. That is, the switching transistor 76e is connected between the node 762 and the node 701.

The diode 76d is connected in parallel with the one or more switching transistors 76b. The diode 76d is connected between the negative terminal of the direct-current power source 75 and the node 762. The anode of the diode 76d is connected to the negative terminal of the direct-current power source 75, and the cathode of the diode 76d is connected to the node 762.

In the power source system 70B, the pulse controller PC is configured to set an output period of the pulse of the voltage from the pulse unit 76 by supplying a control signal to the pulse unit 76. In an embodiment, the pulse controller PC supplies a pulse control signal as the control signal to the control terminals of the one or more switching transistors 76a and the control terminals of the one or more switching transistors 76b.

Further, in the power source system 70B, the pulse controller PC is configured to supply the pulse control signal to the control terminal of the switching transistor 76e. The switching transistor 76e switches between an open state and a closed state according to the pulse control signal that is supplied from the pulse controller PC to the control terminal thereof.

When the switching transistor 76e is in an open state, the pulse unit 76 is electrically separated from the node 701 and the lower electrode 18. When the switching transistor 76e is

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in a closed state, the pulse unit **76** is connected to the node **701** and the lower electrode **18**.

When the one or more switching transistors **76a** enter a closed state and the one or more switching transistors **76b** enter an open state by the pulse control signal from the pulse controller PC, the node **762** is connected to the ground. As a result, the voltage in the node **762** becomes 0 V (that is, the reference level).

When the one or more switching transistors **76a** enter an open state and the one or more switching transistors **76b** enter a closed state by the pulse control signal from the pulse controller PC, the node **762** is connected to the negative terminal of the direct-current power source **75**.

When the pulse unit **76** is connected to the node **701** and the node **762** is connected to the negative terminal of the direct-current power source **75**, a pulse of a negative voltage (for example, a negative direct-current voltage) is output from the pulse unit **76**. The pulse of the negative voltage that is output from the pulse unit **76** to the lower electrode **18** is used as the other of the first pulse PL1 and the second pulse PL2. In a case where the pulse unit **76** outputs the first pulse PL1, the pulse unit **76** may further output the third pulse PL3.

Hereinafter, FIG. 9 will be referred to. FIG. 9 is a diagram showing a configuration of a substrate support according to another exemplary embodiment. A substrate support **16A** shown in FIG. 9 can be adopted in place of the substrate support **16** in the plasma processing apparatus **1**.

The substrate support **16A** has a lower electrode **18A** and an electrostatic chuck **20A**. The lower electrode **18A** is formed of a conductive material such as aluminum and has a substantially disk shape. The radio frequency power source **61** may be connected to the lower electrode **18A** through the matcher **61m**. As described above, the radio frequency power source **61** may be connected to the upper electrode **30** through the matcher **61m**.

The electrostatic chuck **20A** is provided on the lower electrode **18A**. The electrostatic chuck **20A** has a dielectric portion **20d** and an electrode **21a**. The electrostatic chuck **20A** may further have an electrode **22a** and an electrode **22b**. When the substrate **W** is processed in the internal space **10s**, the substrate **W** is placed on the electrostatic chuck **20A** and held by the electrostatic chuck **20A**. Further, the edge ring ER is mounted on the substrate support **16A**. The edge ring ER is mounted on the substrate support **16A** such that the central axis thereof coincides with the axis AX. The substrate **W** accommodated in the chamber **10** is disposed on the electrostatic chuck **20A** and in the region surrounded by the edge ring ER.

The substrate support **16A** has a first region **21** and a second region **22**. In FIG. 9, the boundary between the first region **21** and the second region **22** is shown by a broken line. The first region **21** is a central region of the substrate support **16A**. The first region **21** includes a central region of the electrostatic chuck **20A** and a central region of the lower electrode **18A**. The second region **22** extends in a circumferential direction on the outside in a radial direction with respect to the first region **21**. The second region **22** includes a peripheral region of the electrostatic chuck **20A** and a peripheral region of the lower electrode **18A**. The electrostatic chuck of the first region **21** and the electrostatic chuck of the second region **22** may be provided by a single electrostatic chuck. That is, the electrostatic chuck of the first region **21** and the electrostatic chuck of the second region **22** may be integrated. In another embodiment, the

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electrostatic chuck of the first region **21** and the electrostatic chuck of the second region **22** may be separate electrostatic chucks.

The first region **21** is configured to support the substrate **W** that is placed thereon (that is, on the upper surface thereof). The first region **21** is a region having a disk shape. The central axis of the first region **21** substantially coincides with the axis AX. The first region **21** shares the dielectric portion **20d** with the second region **22**. The dielectric portion **20d** is formed of a dielectric such as aluminum nitride or aluminum oxide. The dielectric portion **20d** has a substantially disk shape. In an embodiment, the thickness of the dielectric portion **20d** in the second region **22** is smaller than the thickness of the dielectric portion **20d** in the first region **21**. The position in the vertical direction of the upper surface of the dielectric portion **20d** in the second region **22** may be lower than the position in the vertical direction of the upper surface of the dielectric portion **20d** in the first region **21**.

The first region **21** has the electrode **21a** (a chuck electrode). The electrode **21a** is an electrode having a shape of a film and is provided in the dielectric portion **20d** within the first region **21**. A direct-current power source **101p** is connected to the electrode **21a** through a switch **101s**. When the direct-current voltage from the direct-current power source **101p** is applied to the electrode **21a**, an electrostatic attraction force is generated between the first region **21** and the substrate **W**. Due to the generated electrostatic attraction force, the substrate **W** is attracted to the first region **21** and held by the first region **21**.

The first region **21** further has a first electrode **21c**. The first electrode **21c** is an electrode having a shape of a film and is provided in the dielectric portion **20d** within the first region **21**. The electrode **21a** may extend closer to the upper surface of the first region **21** than the first electrode **21c** in the vertical direction.

A power source system **81** is connected to the first electrode **21c** through a filter **81f**. The power source system **81** has the same configuration as the power source system **70** and is configured to apply the bias voltage described above to the first electrode **21c**. The filter **81f** has the same configuration as the filter **70f**.

The second region **22** extends to surround the first region **21**. The second region **22** is a substantially annular region. The central axis of the second region **22** substantially coincides with the axis AX. The second region **22** is configured to support the edge ring ER that is placed thereon (that is, on the upper surface thereof). The second region **22** shares the dielectric portion **20d** with the first region **21**.

In an embodiment, the second region **22** may hold the edge ring ER by an electrostatic attraction force. In this embodiment, the second region **22** may have one or more electrodes (chuck electrodes). In the example shown in FIG. 9, the second region **22** has a pair of electrodes, that is, the electrodes **22a** and **22b**. The electrodes **22a** and **22b** are provided in the dielectric portion **20d** within the second region **22**. The electrodes **22a** and **22b** configure a bipolar electrode. Each of the electrodes **22a** and **22b** is an electrode having a shape of a film. The electrodes **22a** and **22b** may extend at substantially the same height position in the vertical direction.

A direct-current power source **102p** is connected to the electrode **22a** through a switch **102s** and a filter **102f**. The filter **102f** is configured to cut off or attenuate radio frequency power. A direct-current power source **103p** is connected to the electrode **22b** through a switch **103s** and a filter **103f**. The filter **103f** is configured to cut off or reduce radio frequency power.

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The direct-current power source **102p** and the direct-current power source **103p** apply a direct-current voltage to the electrode **22a** and the electrode **22b**, respectively, in order to generate an electrostatic attraction force that attracts the edge ring ER to the second region **22**. The set potential of each of the electrodes **22a** and **22b** may be any of a positive potential, a negative potential, and 0 V. For example, the potential of the electrode **22a** may be set to a positive potential, and the potential of the electrode **22b** may be set to a negative potential. Further, the potential difference between the electrodes **22a** and **22b** may be formed by using a single direct-current power source, instead of two direct-current power sources.

When direct-current voltages are applied to the electrodes **22a** and **22b**, an electrostatic attraction force is generated between the second region **22** and the edge ring ER. The edge ring ER is attracted to the second region **22** by the generated electrostatic attraction force and is held by the second region **22**.

The second region **22** further has a second electrode **22c**. The second electrode **22c** is an electrode having a shape of a film. The second electrode **22c** is provided in the dielectric portion **20d** within the second region **22**. The second electrode **22c** is separated from the first electrode **21c**. The electrodes **22a** and **22b** may extend closer to the upper surface of the second region **22** than the second electrode **22c** in the vertical direction. The second electrode **22c** may be disposed outside the second region **22**. For example, the second electrode **22c** may be provided below the edge ring ER and in the insulating part **29**.

A power source system **82** is connected to the second electrode **22c** through a filter **82f**. The power source system **82** has the same configuration as the power source system **70** and is configured to apply the bias voltage described above to the second electrode **22c**. The filter **82f** has the same configuration as the filter **70f**. A bias voltage that is applied to the second electrode **22c** by the power source system **82** may be synchronized with a bias voltage that is applied to the first electrode **21c** by the power source system **81**. The phase of the bias voltage that is applied to the second electrode **22c** by the power source system **82** may coincide with the phase of the bias voltage that is applied to the first electrode **21c** by the power source system **81**. The bias voltage that is applied to the second electrode **22c** by the power source system **82** may have a phase difference with respect to the bias voltage that is applied to the first electrode **21c** by the power source system **81**.

The second region **22** may further have a gas line **22g**. The gas line **22g** is a gas line provided to supply a heat transfer gas, for example, a He gas, to the gap between the second region **22** and the edge ring ER. The gas line **22g** is connected to a gas supply mechanism **104** which is a source for the heat transfer gas.

The power source system **81** may be connected to the electrode **21a**. In this case, the substrate support **16A** does not need to have the first electrode **21c**. Further, the power source system **82** may be connected to at least one of the electrode **22a** and the electrode **22b**. In this case, the substrate support **16A** does not need to have the second electrode **22c**. Further, the second region **22** does not need to have the electrode **22a** and the electrode **22b**. Further, the power source system **81** may apply a bias voltage to the first electrode **21c** and the second electrode **22c**. In this case, the plasma processing apparatus **1** does not need to be provided with the power source system **82**.

Further, the electrostatic chuck **20A** may have a source electrode to which the radio frequency power from the radio

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frequency power source **61** is supplied. The source electrode is provided in the dielectric portions **20d** of at least one of the first region **21** and the second region **22**. In a case where the source electrode is provided in the dielectric portion **20d** of the first region **21**, the source electrode may be provided below the first electrode **21c**, and the first electrode **21c** may be provided below the electrode **21a**. Alternatively, the source electrode may be provided above the first electrode **21c** and below the electrode **21a**. In a case where the source electrode is provided in the dielectric portion **20d** of the second region **22**, the source electrode may be provided below the second electrode **22c**, and the second electrode **22c** may be provided below the electrodes **22a** and **22b**. Alternatively, the source electrode may be provided above the second electrode **22c** and below the electrodes **22a** and **22b**.

While various exemplary embodiments have been described above, various additions, omissions, substitutions and changes may be made without being limited to the exemplary embodiments described above. Elements of the different embodiments may be combined to form another embodiment.

For example, each of the power source systems **70**, **81**, and **82** may have one or more power sources (for example, direct-current power sources) capable of outputting a pulse of a voltage (for example, a direct-current voltage), instead of the configurations shown in FIGS. 7 and 8. Each of the one or more power sources may be a variable direct-current power source. The bias voltages of various examples described above can be generated by such one or more power sources.

Further, the pulse controller PC may be a configuration associated with the radio frequency power source **61** or the radio frequency power source **62**, instead of the configurations of the power source systems **70**, **81**, and **82**.

Further, in the second example shown in FIG. 3, the second pulse PL2 may be generated by connecting the node **722** to the ground after the point in time of end of the first period P1, and electrically separating the node **722** from the direct-current power source **71** before the potential of the node **722** becomes 0 V. In order to connect the node **722** to the ground, the one or more switching transistors **72a** are set to a closed state and the one or more switching transistors **72b** are set to an open state. In order to electrically separate the node **722** from the direct-current power source **71**, the one or more switching transistors **72a** and the one or more switching transistors **72b** are set to an open state.

Further, in another embodiment, the plasma processing apparatus may be another type of plasma processing apparatus, instead of the capacitively coupled plasma processing apparatus. The other type of plasma processing apparatus may be an inductively coupled plasma processing apparatus, an electron cyclotron resonance (ECR) plasma processing apparatus, or a plasma processing apparatus that generates plasma by using surface waves such as microwaves. In the plasma processing methods according to various exemplary embodiments, such other types of plasma processing apparatuses may be used.

From the foregoing description, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

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What is claimed is:

1. A plasma processing apparatus comprising:

a chamber;

a substrate support having an electrode and disposed in the chamber; and

a power source system configured to apply a pulsed voltage signal to the electrode in cycles, wherein each cycle consists of

a sequence of first voltage pulses, each of which has a first voltage level and a negative polarity, in a first period in each cycle,

a sequence of second voltage pulses, each of which has a second voltage level different from the first voltage level and a negative polarity, in a second period,

a third voltage pulse having a positive polarity in a third period between the first period and the second period, the third period being a single period between the first period and the second period, and the third voltage pulse being a single voltage pulse in the third period, and

a fourth voltage pulse having a positive polarity in a fourth period after the second period, the fourth voltage pulse being a single voltage pulse in the fourth period.

2. The plasma processing apparatus according to claim 1, wherein the power source system is configured to periodically output the first voltage pulses at a frequency which is not lower than 100 kHz and not higher than 1 MHz in the first period, and periodically output the second voltage pulses at a frequency which is not lower than 100 kHz and not higher than 1 MHz in the second period.

3. The plasma processing apparatus according to claim 1, wherein a frequency which is a reciprocal of a time length of the cycle is not lower than 0.2 Hz and not higher than 1 Hz.

4. The plasma processing apparatus according to claim 1, wherein the power source system is configured to set an absolute value of the first voltage level to be larger than an absolute value of the second voltage level.

5. The plasma processing apparatus according to claim 1, wherein the power source system is configured to set an absolute value of the first voltage level to be smaller than an absolute value of the second voltage level.

6. The plasma processing apparatus according to claim 1, wherein

the substrate support is configured to support a substrate, and

the power source system is configured to apply the pulsed voltage signal to the electrode to draw ions from a plasma in the chamber into the substrate on the substrate support.

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7. A plasma processing method comprising:

preparing a substrate on a substrate support provided in a chamber of a plasma processing apparatus, wherein the substrate support includes an electrode; and

outputting a pulsed voltage signal from a power source system to the electrode in cycles, wherein each cycle consists of

a sequence of first voltage pulses, each of which has a first voltage level and a negative polarity, in a first period,

a sequence of second voltage pulses, each of which has a second voltage level different from the first voltage level and a negative polarity, in a second period,

a third voltage pulse having a positive polarity in a third period between the first period and the second period, the third period being a single period between the first period and the second period, and the third voltage pulse being a single voltage pulse in the third period, and

a fourth voltage pulse having a positive polarity in a fourth period after the second period, the fourth voltage pulse being a single voltage pulse in the fourth period.

8. The plasma processing method according to claim 7, wherein the first voltage pulses are periodically output at a frequency which is not lower than 100 kHz and not higher than 1 MHz in the first period, and the second voltage pulses are periodically output at a frequency which is not lower than 100 kHz and not higher than 1 MHz in the second period.

9. The plasma processing method according to claim 7, wherein a frequency which is a reciprocal of a time length of the cycle is not lower than 0.2 Hz and not higher than 1 Hz.

10. The plasma processing method according to claim 7, wherein an absolute value of the first voltage level is set to be larger than an absolute value of the second voltage level.

11. The plasma processing method according to claim 7, wherein an absolute value of the first voltage level is set to be smaller than an absolute value of the second voltage level.

12. The plasma processing method according to claim 7, wherein

the substrate support is configured to support a substrate, and

the pulsed voltage signal is applied to the electrode to draw ions from a plasma in the chamber into the substrate on the substrate support.

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